

ASSESSMENT TOOL – 3

Course Code	CSA0702	Course Title	Computer Networks
CO Assessed	CO4	Assessment	Comparative Analysis
Weightage		Total Marks	

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Faculty Incharge	Dr R S Selvan	Assessment	Individual Submission

Total Marks		Obtained Marks	
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Code of Conduct:

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I understand that any violation of academic integrity rules will result in disciplinary action.

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Comparative Analysis

Star vs Bus Topology - Cost, Reliability and Performance

Star Topology is generally more expensive than bus topology because it requires a central switch or hub and separate cables for each node. Bus topology uses a single backbone cable, making it cheaper and easier to install. However, star topology is more reliable because the failure of one node does not affect others, while a backbone failure can stop the entire bus network. Star topology also provides better performance because the central device manages data transmission and reduces collision.

2 Mesh vs Star Topology - Fault Tolerance

Mesh topology provides very high fault tolerance because each node has multiple connections and alternative paths are available for data transmission. If one link fails, communication can continue through another path. Star topology has moderate fault tolerance because individual node failures do not affect the network, but failure of the central switch or hub can stop the entire network. Therefore, mesh topology is more robust for critical applications.

3 Bus vs Mesh topology - Installation Complexity

Bus Topology is simple and easy to install

because it uses a single backbone cable with fewer connections. Mesh topology is much more complex because each node failure does not affect the network, but failure of the central switch or hub can stop the entire network. Therefore, mesh topology is more robust for critical applications.

4 Hybrid vs Star topology - Scalability

Hybrid topology offers high scalability because it combines two or more topologies, such as star, bus, or mesh. New network segments can be added according to organizational requirements without significantly affecting existing sections. Star topology is also scalable, but its expansion depends on the available ports and capacity of the central switch or hub. Hence, hybrid topology provides greater flexibility for large networks.

5 Star, Mesh, Bus and Hybrid Topologies - Maintenance

Star topology is relatively easy to maintain because faults can be isolated to individual nodes or cables. Mesh topology is difficult to maintain due to its large number of connections, although it provides high reliability. Bus topology is also harder to troubleshoot because a fault in the backbone cable can affect the entire network. Hybrid topology has moderate maintenance complexity because managing multiple combined topologies requires careful monitoring and skilled administration.

Network Topologies

Star Topology

- * Cost
- * Reliable
- * Good Performance
- * Easy maintenance

Bus Topology

- * Low cost
- * Simple
- * Easy setup
- * Backbone failure risk

Mesh Topology

- * High cost
- * High fault tolerance
- * Complex setup
- * Highly reliable

Hybrid Topology

- * Highly scalable
- * Flexible
- * Combines multiple topologies
- * Moderate maintenance

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